

AMERICAN UNIVERSITY OF PHNOM PENH

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COSC 251: Data Structures – Group Project

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Project Proposal Deadline: November 2, 2026
Final Project Submission Deadline: November 30, 2026

Project Purpose

The purpose of this project is to apply your knowledge of data structures, algorithms, and computational complexity to solve a challenging computational problem.

Working in groups, you will select and research a computational problem, study one or more algorithmic solutions, implement a working program, evaluate its performance, and present your findings to the class.

By completing this project, you should be able to:

- Understand and explain a computational problem.
- Identify appropriate algorithms and data structures.
- Analyze the correctness, efficiency, and limitations of an algorithm.
- Implement and test an algorithm using a programming language.
- Evaluate performance using different input sizes.
- Communicate technical concepts and findings clearly.

Project Requirements

1. **Group Work**

   This is a group project. All group members are expected to contribute meaningfully to the research, implementation, testing, report, and presentation.

   Group members should divide responsibilities fairly and collaborate throughout the project.

   All members must understand the overall project, not only the part assigned to them.

2. **Topic Selection and Proposal**

   Your group must select an interesting computational problem that is classified as **NP-Complete or NP-Hard**.

   The project must focus on:

   - A clearly defined computational problem.

- One or more algorithms or data structures used to solve or approximate the problem.
- An explanation of why the problem is computationally challenging.

Your selected topic must be approved by the instructor before beginning the full research and implementation.

Examples of possible topics include:

- Traveling Salesman Problem (TSP)
- 0/1 Knapsack Problem
- Graph Coloring Problem
- Sudoku Solving
- Set Cover Problem
- Bin Packing Problem
- Job Shop Scheduling Problem
- Hamiltonian Cycle Problem
- Subset Sum Problem

These are examples only. You may propose another computational problem, subject to instructor approval.

Some problems have different complexity classifications depending on whether they are formulated as decision or optimization problems. Your group should clearly identify the version being studied.

Proposal Submission

Submit a short proposal on Telegram Group containing:

- **Title:** The proposed project title.
- **Problem:** A brief description of the computational problem and its importance.
- **Solution:** A brief description of the proposed algorithm(s) or data structure(s).
- **Abstract:** One short paragraph describing your planned research, implementation, testing, report, and presentation.

Proposal Deadline: November 2, 2026

3. **Research and Study**

Your group must research and study the selected computational problem and algorithmic solution.

Your research should include:

- A clear definition and explanation of the computational problem.
- Examples illustrating how the problem works.
- Real-world applications of the problem.
- An explanation of its computational complexity classification.
- A description of one or more possible algorithmic solutions.
- An explanation of the data structures used by the selected algorithm.
- An analysis of the algorithm, including its main operations, time complexity, advantages, and limitations.

Your group must also explain whether the selected algorithm produces:

- An **exact solution**, which guarantees an optimal answer for an optimization problem; or

- An **approximate or heuristic solution**, which may not guarantee an optimal answer.

If your algorithm does not guarantee an optimal solution, explain its limitations and how you will evaluate the quality of its results.

You are not required to implement every algorithm you research. However, you must understand and implement at least one appropriate solution.

4. **Implementation and Demonstration**

   Your group must implement the selected algorithmic solution in a programming language of your choice.

   The implementation must satisfy the following:

   - It must solve the selected computational problem using the algorithm studied.
   - The core algorithm must be implemented by your group rather than relying entirely on a library that already solves the problem.
   - The source code must be clearly organized, readable, and maintainable.
   - It must allow the instructor to test the program using custom input, where appropriate.
   - It must include clear instructions explaining how to install, compile, and run the program.
   - It must demonstrate the solution using small and larger test cases.
   - It must display or report the solution produced by the algorithm.
   - It must measure and report execution time for different input sizes.
   - It must explain any limitations encountered when solving larger instances.

   Testing Requirements

   Your group must test the implementation using multiple input instances.

   At minimum, your testing should include:

   (a) **Small Test Case:** Use a small input where the expected result can be verified manually or independently.

   (b) **Larger Test Case:** Use a larger input to demonstrate the algorithm's performance and limitations.

   (c) **Performance Analysis:** Measure execution time and explain how performance changes as input size increases.

   You are encouraged to test additional input sizes to make your performance analysis more meaningful.

   The appropriate input size depends on the selected problem and algorithm. A larger test case does not need to contain millions of elements.

   For approximate or heuristic algorithms, you should also evaluate solution quality where possible, using known optimal solutions, bounds, or other reasonable comparisons.

   Your program must be ready for a live demonstration. The instructor may provide custom input to test your implementation.

5. **Research Report**

   Submit a short research report in PDF format.

   The recommended length is approximately **4–6 pages**, excluding references and appendices.

   The report must contain the following sections:

a. **Problem**

Define and explain the selected computational problem.

Include:

- Problem definition and examples.
- Real-world applications.
- Computational complexity classification.
- Why the problem is challenging.

b. **Solution**

Explain the selected algorithm and data structures used to solve the problem.

Include:

- How the algorithm works.
- Important steps or operations.
- Time complexity analysis.
- Advantages and limitations.
- Whether the solution is exact, approximate, or heuristic.

c. **Implementation**

Describe how your group implemented the algorithm.

Include:

- Programming language used.
- Important components of the program.
- Data structures used.
- How to run and use the program.

d. **Findings**

Present the results of your implementation and testing.

Include:

- Test cases and results.
- Execution time for different input sizes.
- Observations about algorithm performance.
- Solution quality, where applicable.
- Limitations and challenges encountered.
- Conclusions from your findings.

You may include diagrams, pseudocode, screenshots, tables, graphs, or other supporting materials.

All external sources, algorithms, diagrams, and references must be properly cited.

6. **Presentation**

Prepare a presentation based on your research, implementation, and findings.

The presentation should be approximately **15 minutes**, followed by approximately **5 minutes** for questions and answers.

Your presentation should cover:

(a) **Problem Introduction**

Explain the computational problem, why it is important, and where it is used.

(b) **Algorithm and Data Structures**

Explain the selected solution, how the algorithm works, and which data structures are used.

Discuss its time complexity, advantages, and limitations.

(c) **Implementation and Demonstration**
Demonstrate your working program.
Show how the program processes input and produces a solution.

(d) **Testing and Findings**
Present your test cases, execution times, performance observations, and conclusions.

All group members are expected to participate in the presentation.

Slides must be written in your own words. Any external sources, images, diagrams, or references must be properly cited.

7. **Individual Understanding**

Every group member is expected to understand:

- The computational problem.
- The selected algorithm and data structures.
- How the algorithm works.
- The overall implementation.
- The time complexity and limitations.
- The testing results and findings.

You are not required to memorize every line of code.

However, you must be able to explain how your solution works and why the selected approach is appropriate.

During the presentation and demonstration, the instructor may ask any group member questions about the project.

Individual grades may differ depending on each student's demonstrated understanding and contribution.

Submission Requirements

Submit the following items on Canvas by **November 30, 2026**:

1. **Presentation Slides**

Submit the slides used for your presentation.

2. **Research Report**

Submit your research report in PDF format.

3. **Source Code**

Submit the complete source code of your implementation, including any necessary input files or test cases.

4. **README / Running Instructions**

Include a README file explaining:

- Programming language and version.
- Required dependencies, if any.
- Installation or compilation steps.

- Instructions for running the program.
- How to provide custom input.
- Example input and expected output.

The source code and supporting files should be organized into a single ZIP file.

The submitted program must be runnable using the provided instructions.

Grading Criteria

The project will be graded according to the following criteria:

Component	**Points**
Problem Definition and Research	15
Algorithm and Data Structure Explanation	20
Implementation and Correctness	25
Testing and Performance Analysis	15
Research Report	10
Presentation and Demonstration	10
Individual Understanding	5
Total	**100**

Note: Individual grades may differ based on each student's contribution, participation, and demonstrated understanding.